User Manual for THEIA

User Manual

Version 1.0, Draft 2

Vijay Deva
Aryan Puthran
Julia Smith (Liaison)

Washington State University - CPTS 484

09/30/2026

Table of Contents

Table of Contents ii
Revision History iii
1. Overview 1
    1.1 Project Purpose 1
    1.2 Quick Start Checklist 1
2. Getting Started 2
    2.1 Interface Walkthrough 2
3. Core Features & Workflows 3
    3.1 Initial Room Wayfinding 3
    3.2 Precise Indoor Turn Navigation 3
    3.3 Hazard and Obstacle Avoidance 3
    3.4 Low Vision Visual Accessibility 3
    3.5 Automated Emergency Response 4
    3.6 Advanced Navigation Settings 4
4. Troubleshooting 5
    4.1 Common Error Messages 5

Revision History

Name	Date	Reason for Changes	Version
Initial Draft	09/30/2026	Initial draft.	1.0, Draft 1
Final PP 1 Report	10/15/2026	Updated for our FPP1 submission.	1.0, Draft 2

1. Overview

1.1. Project Purpose

THEIA is a smartphone application designed to assist blind and visually impaired individuals with indoor navigation. The application guides users from one indoor location to another across connected buildings. The system ensures safe and fast indoor navigation relying on voice and audio feedback instead of screen readability.

1.2. Quick Start Checklist

- Open the preferences panel to adjust voice volume, speech rate, text size, and distance units.
- Add trusted family members and local emergency services to the emergency contacts profile.
- Download offline maps in the cache manager for reliable navigation in buildings when network connections drop.
- Walk a known distance using the optional step calibration utility to personalize your distance cues.

2. Getting Started

2.1. Interface Walkthrough

The application interface provides a high contrast, large text visual display alongside audio and haptic feedback. The visual interface uses a contrast ratio of at least 7:1 and text scalable up to 200 percent. Users can operate the app using the voice command query processor to search for rooms naturally. Users can also tap frequent destination cards directly on the screen using large tactile touch areas.

3. Core Features & Workflows

3.1. Initial Room Wayfinding

- Users can initiate a voice search or select from a high contrast tactile list of frequent indoor destinations.
- Tap the voice input button to activate the query processor.
- Speak a room name, place, or facility type.
- The application displays the transcribed destination and prompts the user to confirm the location before routing begins.
- When approaching from outside, the entrance transition indicator notifies the user that they have reached the building entrance.
- The system then confirms that it is automatically switching to indoor navigation.
- The app presents a route selection card displaying multiple calculated indoor paths.
- Users can select a comfortable stair free route that utilizes elevators or choose the shortest path involving stairs.

3.2. Precise Indoor Turn Navigation

- The turn-by-turn guidance display shows the active indoor route with a large directional arrow.
- The interface provides current distance readouts in feet and previews the next upcoming turn.
- The application delivers real-time voice cues instructing the user to walk forward a set distance.
- The system prompts the user at the exact moment they need to execute a turn.
- Users can tap the high contrast Repeat Instruction button at any point to hear the latest audio prompt again.

3.3. Hazard and Obstacle Avoidance

- The application pauses active navigation to display a red alert regarding a floor level hazard detected ahead.
- The screen provides a clear reminder that overhead hazards are not detected.
- The system instructs the user to stop and follow the audio guidance around the obstacle when it is safe.
- A maintenance reroute notification alerts the user to unexpected pathway closures caused by wall maintenance.
- The interface immediately provides a new accessible detour path, details the added travel time, and prompts the user to continue on the new route.

3.4. Low Vision Visual Accessibility

- The application displays an interface with high contrast bold visuals and magnified text to assist users with low vision.
- Visual directional markers use bold yellow shapes on a deep black background to minimize eye strain and disorientation.
- Users can open the preferences panel to customize the visual experience by scaling the text size up to 200 percent.
- The preferences panel also allows users to toggle the screen contrast mode and adjust distance units.

3.5. Automated Emergency Response

- If the smartphone accelerometer registers a fall impact, an emergency alert modal triggers and initiates a ten second haptic countdown.
- The user can cancel the automated emergency dispatch using a voice command or by tapping the designated cancellation button.
- If the user does not respond within thirty seconds, the app sends an emergency distress signal containing their indoor coordinates to campus security.
- Users can navigate to the safety profile to select trusted family members and local emergency services to receive these distress alerts.
- If indoor tracking confidence drops to an unreliable level, a signal reacquisition screen directs the user to stop walking immediately.
- The system stays in place and attempts to reconnect while providing a quick option to call for help.

3.6. Advanced Navigation Settings

- Users can adjust voice volume, speech rate, and instruction frequency within the preferences panel to tailor audio feedback to their environment.
- The offline map cache manager allows users to download building floor plans to support indoor guidance when network reliability drops.
- The offline manager displays a warning that maintenance closures may be out of date without an active connection.
- The optional step calibration utility allows users to walk a familiar path to calibrate their personal step length.
- This calibration adds a personal estimate based on the user’s usual pace as the primary distance cue instead of standard feet or meters.

4. Troubleshooting

4.1. Common Error Messages

- Location Lost: The screen displays a Please STOP alert when indoor tracking drops. Stand still while the app listens for nearby connections to re-establish your coordinates.
- Offline Limitation: The offline map manager displays a warning that maintenance closures may be out of date when using the app without an active network connection. Connect to a network to sync the latest facility closure updates.
- Obstacle Detection Limitation: During an active route or when a hazard is detected, a persistent warning banner reminds users that the app detects floor level obstacles only. The system will explicitly state that overhead hazards are not detected, meaning users must continue using a white cane for full body protection.
- Voice Recognition Failure: If the app misunderstands a spoken destination due to background noise, the voice command query processor displays exactly what it heard. You can press the Try again button to restart the listening prompt and repeat your destination.
- Stair Free Route Unavailable: If you request a comfortable route but the building lacks elevators or ramps, the system will announce that no stair free route exists. The app will then ask you how you wish to proceed.
- Calibration Safety Warning: When opening the optional step calibration utility, the screen displays a red safety warning. This alert reminds users to only start calibration in known locations to ensure physical safety while the app calculates regular walking pace.